

## Intel and Nokia combine?

The two giants are rumoured to be working on a Atom-based SoC for cellphones codenamed Penwell

## Notebooks with Radeon 5870s

ASUS has gone ahead and offered a notebook solution with a Radeon HD 5870 GPU, beating NVIDIA's GTX280 solution

# Mean and Green

**Better late than never is the adage. We say it had better be better. NVIDIA's upcoming DX11 part gets dissected**

Michael Browne

[michael.browne@thinkdigit.com](mailto:michael.browne@thinkdigit.com)

**R**ound one of the DX11 battle is over...AMD has won, and NVIDIA goes home without a runners-up trophy. Writing off AMD as "all washed up" was foolhardy and this coupled with NVIDIA's 40nm problems spelt disaster for them during the age of DX11's infancy. AMD has sold many DX11 parts, thereby restoring coffers sans competition.

NVIDIA claims their 40nm woes "are history", and our monolithic protagonist is in production as you read this. The GTX 480 and 470 based on the GF100 have been announced. But Fermi's adversary has a running start and DX 11 has reached proving point with people already tapping their feet impatiently. There is a lot riding on this one, despite the apparent air of nonchalance NVIDIA manages to exude.

## The GF100 exposed

Fermi is huge – at 3 billion transistors, it's more than twice of the GT200s (GTX 280) count of 1.4 billion. 512 stream processors (SPs), a 384-bit GDDR5 interface – Fermi is most impressive on paper, much like our cricket team! But is it more than two GT200s on a single die?

Fermi is scalable. Thanks to decoupling of functional units on the GPU, lower-end chips will have the same logic as top-end ones. The 512SPs are divided into 16 groups of Streaming Multiprocessors (SMs). Furthermore, one cluster of four SMs makes up one Graphic Processing Cluster (GPC).



The GF100 architecture in detail

The Host Interface is an interconnect between the CPU and GPU, while the GigaThread Engine, which is basically the scheduler, is responsible for fetching and copying data from system RAM to the video RAM via the six memory controllers. This GigaThread Engine also handles redistribution of any workload.

Each memory controller is 64-bits wide, which is how NVIDIA arrives at the figure "384-bit" wide controller. Blocks of threaded instructions are created within the GigaThread Engine and sent off to the SMs. Each SM then breaks down this block and feeds it, 32 threads at a time

to the SPs within the block, each SP getting one thread. Remember that each SP can execute pixel, vertex, geometry and

compute instructions. The large block of common L2 cache (768 KB) handles load, store and texture operations. The 48 ROP units are divided into six clusters of eight units each, and each cluster gets its very own 64-bit memory controller. As you can see, bus bandwidth and the ROP units are very important to NVIDIA, which is why these are very close to the L2 cache and memory controller. We're also told scaling on each of these units is linked, meaning changing the clocks of one, will affect the other units. On the GT200 and previous generation GPUs, SMs and Texture Units were grouped together and collectively referred to as Texture Processing Clusters, whereas on the GF100, each SM has four texture units (shaded in dark blue) – more of the scalar concept. The SP has a new trick in its ability to perform a two-step multiply-add operation in one step with a new instruction called FMA

CAN YOU DESIGN A BETTER LOGO THAN THIS?



Amit Ranjan Srivastava

Visit our Facebook Fan page:  
[facebook.com/thinkdigit](https://www.facebook.com/thinkdigit) and

**WIN EXCITING PRIZES!**